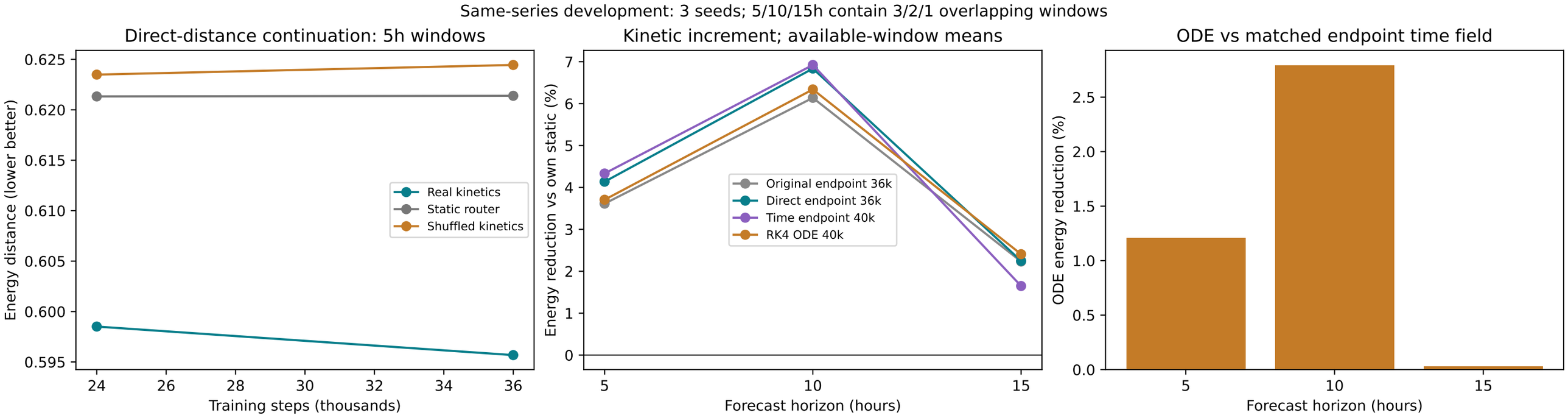
Same-series development: 3 seeds; 5/10/15h contain 3/2/1 overlapping windows
Direct-distance continuation: 5h windows
Energy distance (lower better)
0.625
0.620
0.615
0.610
0.605
0.600
0.595
Real kinetics
Static router
Shuffled kinetics
24
26
28
30
32
34
36
Training steps (thousands)
Kinetic increment; available-window means
Energy reduction vs own static (%)
7
6
5
4
3
2
1
0
Original endpoint 36k
Direct endpoint 36k
Time endpoint 40k
RK4 ODE 40k
5
10
15
Forecast horizon (hours)
ODE vs matched endpoint time field
ODE energy reduction (%)
2.5
2.0
1.5
1.0
0.5
0.0
5
10
15
Forecast horizon (hours)